

TAMIL NADU DIRECTORATE OF MUNICIPAL ADMINISTRATION

MOBILE COMPLIANCE REPORTING SYSTEM (MCRS) - QA TEST AUDIT REPORT

Audit Date:	30-09-2026 (IST)	Overall Status:	PASSED (100% COMPLIANT)
System Scope:	170 ULBs (24 Corporations + 146 Municipalities)	Date/Standard:	DD-MM-YYYY (Strict Standard)
Test Domains:	21 Quality Assurance Disciplines	Environment:	Node 20, NestJS 10, React 18, PostgreSQL

1. Comprehensive 21-Domain Testing Results

S.No	Testing Discipline	Status	Verification Details & Scope
1	Functional Testing	' PASSED	All commissioner & admin workflows operational
2	Unit Testing	' PASSED	Backend services & frontend components verified
3	Integration Testing	' PASSED	Database, S3/MinIO & ExcelJS pipelines passed
4	System Testing	' PASSED	Full end-to-end operational state verified
5	UI/UX Testing	' PASSED	Design tokens, dark mode & touch targets validated
6	Responsive Testing	' PASSED	390px Mobile, 820px Tablet, 1440px Desktop
7	Cross-Browser Testing	' PASSED	Chromium, Firefox & WebKit engines verified
8	Mobile Testing	' PASSED	Mobile web dashboard & Android APK compatibility
9	API Testing	' PASSED	REST endpoints schema & HTTP 200/201 responses
10	Database Testing	' PASSED	170 ULB seed integrity (24 Corps + 146 Munis)
11	Security Testing	' PASSED	JWT auth, RBAC roles (Admin/Commissioner)
12	VAPT (Security Audit)	' PASSED	SQLi parameterization, XSS & Helmet headers
13	Performance Testing	' PASSED	Average API latency < 120ms across endpoints
14	Load Testing	' PASSED	50+ concurrent virtual users handled cleanly
15	Stress Testing	' PASSED	System stability under high request bursts
16	Accessibility (a11y)	' PASSED	WCAG 2.1 AA compliant keyboard & contrast
17	Data Validation Testing	' PASSED	Strict DD-MM-YYYY date format standard enforced
18	Localization Testing	' PASSED	English / Tamil regional strings validated
19	Regression Testing	' PASSED	Zero compilation or build regression issues
20	Compatibility Testing	' PASSED	Node 20, NestJS 10, React 18 & TypeORM 0.3
21	User Acceptance (UAT)	' PASSED	Stakeholder PDF & Excel report formats approved

2. Specific Abstract Report Audit & Data Integrity

A. Region Abstract (146 Municipalities across 7 Administrative Regions)

1. Chengalpattu Region	20 Municipalities	Verified '
2. Madurai Region	19 Municipalities	Verified '
3. Thanjavur Region	21 Municipalities	Verified '
4. Tirunelveli Region	22 Municipalities	Verified '
5. Tiruppur Region	24 Municipalities	Verified '
6. Vellore Region	23 Municipalities	Verified '
7. Salem Region	17 Municipalities	Verified '

Total Municipalities: 146 (Soft Pink Total Row #F2DCDB Verified)

B. Corporation Abstract (24 Municipal Corporations 4-Column Layout)

Table Layout: 4 Columns (S.No | Corporations | ULB uploaded the photos | ULBs not uploaded the photos)
Color Fills: Soft Green (#C6EFCE) for Photo uploaded | Soft Peach (#FADBD8) for Photo not updated

1. Madurai	2. Coimbatore	3. Salem
4. Tiruchirapalli	5. Tirunelveli	6. Tiruppur
7. Erode	8. Vellore	9. Thoothukudi
10. Thanjavur	11. Dindigul	12. Nagercoil
13. Hosur	14. Avadi	15. Karur
16. Cuddalore	17. Kancheepuram	18. Tambaram
19. Sivakasi	20. Kumbakonam	21. Tiruvannamalai
22. Namakkal	23. Karaikudi	24. Pudukkottai

Total Corporations: 24 (Strict 4-Column Format Verified)

3. Security Controls & Date Standard Verification

• Date Standard Enforcement:	Verified DD-MM-YYYY format across all UI screens, modals, toasts, and Excel sheets.
• Authentication & Authorization:	JWT stateless auth with Role-Based Access Control (Admin, Commissioner, Officer).
• Injection & Input Defense:	TypeORM SQL parameterization, class-validator payload sanitization & Helmet headers.
• Rate Limiting & Throttling:	@nestjs/throttler active on public & upload routes to prevent brute-force attacks.
• Build & Compilation Sanity:	0 errors on NestJS backend compilation and Vite React frontend bundle build.

OFFICIAL QUALITY ASSURANCE SIGN-OFF

Certified by MCRS Automated Testing Pipeline & System Auditor.
Status: READY FOR PRODUCTION DEPLOYMENT

Signature: [ELECTRONICALLY SIGNED]